



1. Identification of the substance/preparation and of the company undertaking

Trade Name: KPL Zinc Oxide

Application: Used in rubber, paints, ceramics, plastics, and as a feed additive or UV absorber

Manufacturer/Supplier:

Kesar Petroproducts Ltd.

Plot No. D-7/1, Lote Parshuram MIDC, Tal-Khed, Dist. Ratnagiri, Maharashtra – 415722

Further Information: Product Safety Department, Kesar Petroproducts Ltd., Ratnagiri Plant

2. Hazards Identification

GHS Classification:

Not classified as flammable, corrosive, or acutely toxic

May cause respiratory irritation (H335: May cause respiratory irritation) if inhaled as fine dust

Label Elements:

Pictogram: None required under current EU/OSHA rules for standard industrial grades

Precautionary Statement: P261 – Avoid breathing dust; P271 – Use only outdoors or in well-ventilated area

3. Composition/Information on Ingredients

Description: Zinc Oxide

Chemical Formula: ZnO

CAS No.: 1314-13-2

Purity: ≥99% (typical industrial grade)

Impurities: Pb < 20 ppm, Cd < 5 ppm, As < 3 ppm (typical)

4. First-Aid Measures

Inhalation: Remove to fresh air. If coughing or shortness of breath occurs, seek medical attention.

Skin Contact: Wash with soap and water.

Eye Contact: Flush with water for 15 minutes. Seek medical advice if irritation persists.

Ingestion: Rinse mouth. Do not induce vomiting. Seek medical attention.

5. Fire-Fighting Measures

Extinguishing Media: Water spray, dry chemical, CO₂

Special Hazards: Emits toxic fumes of zinc oxide when heated strongly ("metal fume fever" risk)

Firefighter Protection: Wear full protective gear and self-contained breathing apparatus

6. Accidental Release Measures

Personal Precautions: Wear dust mask, gloves, and eye protection

Environmental Precautions: Avoid release to soil or water bodies

Cleanup: Dampen to suppress dust. Collect in sealed container. Do not dry sweep.

7. Handling and Storage

Handling: Minimize dust generation. Use local exhaust ventilation. Avoid contact with acids.

Storage: Keep in dry, cool, well-ventilated area in tightly closed containers.

8. Exposure Controls/Personal Protection

Occupational Exposure Limit:

ACGIH TLV-TWA: 10 mg/m³ (total dust); 5 mg/m³ (respirable fraction)

Engineering Controls: Local exhaust ventilation at dust-generating operations

PPE:

Respirator: NIOSH N95 or equivalent if dust > permissible levels

Gloves: Nitrile or neoprene

Eye Protection: Safety goggles

Protective Clothing: Lab coat or coveralls

9. Physical and Chemical Properties

Appearance: White to yellowish-white powder

Odour: Odourless

pH (10% slurry): 7.0 – 8.5

Specific Gravity: 5.6

Melting Point: ~1975°C (sublimes)

Solubility in Water: Insoluble (<0.001 g/100 mL)

Dust Explosibility: Not classified as explosive, but fine dust may pose a nuisance dust hazard

10. Stability and Reactivity

Stable under normal conditions

Incompatible with: Strong acids (releases toxic zinc salts), reducing agents

Decomposition Products: Zinc fumes (toxic) on overheating

11. Toxicological Information

Acute Toxicity (oral, rat): LD₅₀ > 5,000 mg/kg (low toxicity)

Inhalation Hazard: Prolonged high-level exposure to fumes may cause “metal fume fever” (flu-like symptoms)

Skin/Eye Irritation: Low; may cause mild mechanical irritation

Carcinogenicity: Not classified as carcinogenic (IARC Group 3)

12. Ecological Information

Toxicity: Toxic to aquatic life with long-lasting effects (zinc ion release)

Bioaccumulation: Possible in aquatic organisms

Precaution: Prevent environmental discharge

13. Disposal Considerations

Dispose of in accordance with local, state, and national regulations

May be classified as hazardous waste due to zinc content—verify with local authority

Contaminated packaging: Treat as product waste

14. Transportation Information

UN Number: Not regulated

Proper Shipping Name: Zinc oxide

Transport Classification: Not restricted under IMDG, IATA, or ADR

15. Regulatory Information

GHS Status: Not classified as hazardous under CLP (EU) or HCS (USA) for standard grades

REACH: Registered

TSCA: Listed

16. Other Information

This SDS is based on current knowledge and best practices. It does not constitute a guarantee of product performance or safety. Users are responsible for compliance with applicable laws.

Issued by: Quality Control Department, Kesar Petroproducts Ltd.

Date: 29 December 2025